MFJA: Unsupervised Domain Adaptation Based on Multimodal Feature Fusion and Global–Local Joint Alignment for SAR ATR

Chen Zhang, Yinghua Wang, *Senior Member, IEEE*, Hongwei Liu, *Senior Member, IEEE*, Siyuan Wang, Xiaoting Zhang, and Chunhui Qu

***Abstract*— In the field of synthetic aperture radar automatic target recognition (SAR ATR), inherent distributional discrepancies between electromagnetic synthetic and measured SAR images pose significant challenges to the potential applications of the former. To bridge the gap, a novel unsupervised domain adaptation framework based on multimodal feature fusion and global–local joint alignment (MFJA) is proposed in this article. The multimodal feature fusion focuses on describing each target more comprehensively by leveraging both visual and scattering topological information. In the visual branch, the full-aperture image is decomposed into multiple subaperture images to explore the scattering variations in the target at different azimuths, facilitating a richer visual description. Meanwhile, both local scattering and spatial position information of keypoints are simultaneously integrated into the feature extraction in the scattering topological branch, promoting a more comprehensive scattering topological representation. Subsequently, a gated feature fusion module (GFFM) is developed to effectively fuse features derived from different modalities. The global–local joint alignment aims to align different domains with greater precision. Specifically, a power normalized weighted gradient reversal layer (PN-WGRL) is proposed to guide the network to focus more on hard-to-align samples during global domain alignment, thus mitigating their interference with local domain alignment. While MFJA achieves satisfactory cross-domain recognition performance, its inference efficiency is somewhat constrained. Therefore, a domain-invariant cross-modal knowledge distillation (DCKD) algorithm with a tri-path collaborative alignment strategy is further developed to distill discriminative and domain-invariant knowledge from the multimodal model into a compact visual model based on full-aperture images, thereby accelerating inference. Experiments conducted in three scenarios on the public synthetic and measured paired labeled experiment (SAMPLE) dataset validate the effectiveness of both MFJA and DCKD.**

***Index Terms*— Automatic target recognition (ATR), electromagnetic synthetic image, knowledge distillation (KD), multimodal learning, scattering topology, subaperture decomposition, synthetic aperture radar (SAR), unsupervised domain adaptation.**

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NOMENCLATURE

SAR	Synthetic aperture radar.
ATR	Automatic target recognition.
MFJA	Multimodal feature fusion and global–local joint alignment.
SFFN	Subaperture feature fusion network.
GAT	Graph attention network.
PA-GAT	Position-aware GAT.
GFFM	Gated feature fusion module.
GRL	Gradient reversal layer.
WGRL	Weighted GRL.
PN-WGRL	Power normalized WGRL.
LMMD	Local maximum mean discrepancy.
KD	Knowledge distillation.
DCKD	Domain-invariant cross-modal KD.
SimKD	Simple KD.
SAMPLE	Synthetic and measured paired labeled experiment.
ASC	Attribute scattering center.
VSFA	Visual and scattering topological feature fusion and alignment network.
CNN	Convolutional neural network.
SIFT	Scale-invariant feature transform.
SAR-SIFT	SAR-scale-invariant feature transform.
GNN	Graph neural network.
1-D	1-Dimensional.
BN	Batch normalization.
ReLU	Rectified linear unit.
LAFF	Lightweight attention feature fusion.
RKHS	Reproducing kernel Hilbert space.
CORAL	Correlation alignment.
MSE	Mean square error.
SOTA	State-of-the-art.
t-SNE	t-Distributed stochastic neighbor embedding.
GFM	Gated fusion module.
MACs	Multiply–accumulate operations.

I. INTRODUCTION

DIFFERENT from passive optical imaging that captures reflected light from natural or artificial sources to form images and is therefore inherently constrained by environmental conditions such as illumination and atmospheric interference, SAR is an active sensing technology that generates high-resolution imagery by transmitting microwave pulses and coherently processing the backscattering echoes from

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ground targets. Owing to its imaging mechanism, SAR exhibits the capability of round-the-clock and all-weather imaging, making it an important instrument for Earth observation. Recently, ATR has emerged as a crucial area of research within the field of SAR, driven by the increasing demand for intelligent and efficient interpretation of SAR imagery.

Over the past decade, deep learning methods have significantly advanced the field of computer vision. Consequently, an ever-increasing number of researchers have introduced deep learning into SAR ATR, proposing a variety of recognition models with excellent performance [1], [2]. However, acquiring and annotating large amounts of SAR target images are quite difficult in comparison to optical images. To address data scarcity, generative approaches [3], [4] have been explored to synthesize additional training samples, particularly when a small number of measured data are available. For more challenging scenarios where measured data are completely lacking, electromagnetic simulation combined with computer-aided design modeling provides a physically grounded approach [5], [6], [7]. Unfortunately, approximations and simplifications in electromagnetic simulation and model design processes introduce nonnegligible distribution gaps between synthetic and measured SAR images.

To bridge the distribution gap, domain adaptation methods [8] have been widely used in cross-domain recognition tasks based on synthetic SAR images. These algorithms, grounded in distinct assumptions, fall into two principal categories. The first category assumes the availability of a small number of labeled measured images during the training stage (see [6], [7], [9]), while the second category assumes that all the measured images are completely unlabeled (see [10], [11], [12], [13]). The challenging nature of the second assumption motivates its selection for this article.

However, current algorithms suffer from two major limitations.

1) The existing methods rely predominantly on full-aperture SAR amplitude information, while neglecting the unique domain knowledge inherent in SAR imagery. Two representative types of such knowledge are subaperture information, which reveals the target's scattering characteristics at varying azimuths, and scattering topology information, which reflects the target's overall structural properties. Such domain knowledge captures the physical and geometric essence of SAR images, providing insights beyond what can be captured by full-aperture amplitude information alone. It offers dual advantages for cross-domain target recognition. First, the complementary nature of different types of domain knowledge enables a more accurate characterization of the target, thereby facilitating precise distribution alignment across domains under domain adaptation loss constraints. Second, certain types of domain knowledge, such as scattering topology information, exhibit robustness against variations in imaging parameters, inherently mitigating distribution shifts across domains. As shown in Fig. 1, treating SAR images as optical-like inputs in neural networks fails to leverage this valuable domain knowledge, thereby limiting recognition performance.
2) The existing methods primarily focus on global distributional differences at the domain level between synthetic and measured SAR images, while overlooking potential local distributional differences at the class level. As illustrated in Fig. 2, even when domain-level distributions are approximately aligned, some domain shifts may still persist among class-level subdomains, leading to reduced cross-domain recognition performance.

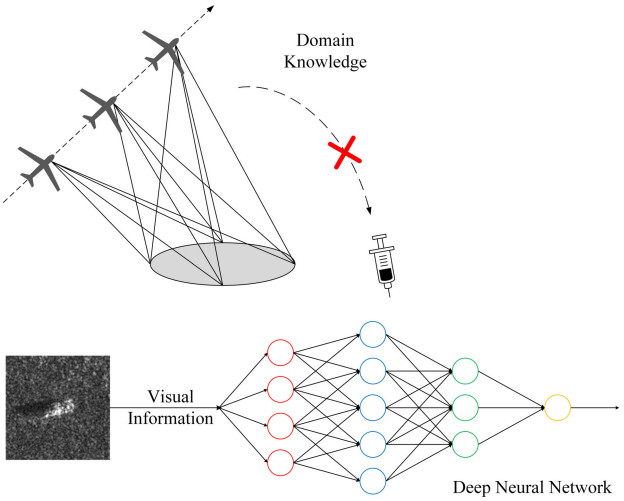


Fig. 1. Illustration of our first motivation. Most algorithms use solely visual information from SAR images, prompting us to explore how to inject SAR domain knowledge into deep learning.

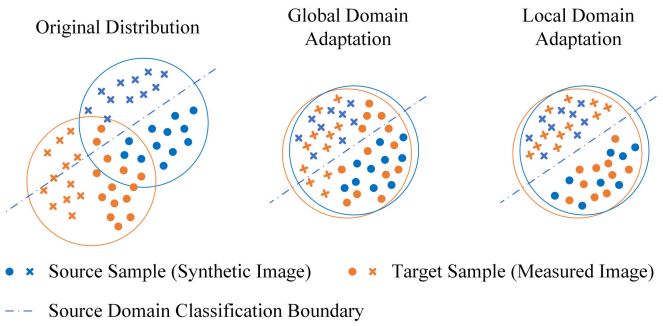


Fig. 2. Illustration of our second motivation. Subdomain shifts may be hidden under the appearance of global alignment, promoting us to reduce the distributional differences between synthetic and measured images at both domain and class levels.

To address the dual limitations of insufficient domain knowledge utilization and coarse alignment, a novel unsupervised domain adaptation framework based on MFJA is proposed in this article. The framework operates through two coordinated mechanisms: 1) domain knowledge-enhanced multimodal learning, where azimuth-dependent scattering characteristics of the target are captured via subaperture decomposition [14] and further fused by an SFFN to enhance visual features. Concurrently, the scattering topological structure of the target is modeled using its keypoints, with both their local scattering and spatial position information fused to enhance topological features via a PA-GAT. A GFFM is then proposed to mitigate intermodal conflicts via learnable

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gates that dynamically recalibrate modality contributions and 2) sample-perceptual joint alignment, featuring our proposed PN-WGRL, which dynamically increases the importance of hard-to-align samples during global domain alignment. This enhanced emphasis helps correct the misalignment of these samples, thereby reducing their detrimental impact on local domain alignment driven by the widely used LMMD loss [15]. These two mechanisms are mutually reinforcing: refined multimodal features provide a more discriminative metric space for domain alignment, and, in turn, the joint alignment process guides the feature extractor to focus on learning domain-invariant representations.

Despite the satisfactory performance gains achieved by leveraging multimodal domain knowledge, subaperture decomposition and scattering topological graph construction inevitably increase inference time, limiting the practical application of MFJA to some extent, especially in inductive scenarios. KD [16] is a widely used technique for model compression. However, most existing methods focus primarily on the discriminative power of knowledge, with few addressing its domain invariance. Even among those that do, the cross-domain constraints are often insufficient.

To bridge the gap between accuracy and efficiency, we further propose a DCKD algorithm to distill discriminative and domain-invariant multimodal knowledge into a unimodal visual model based on full-aperture images, thus providing a lightweight and fast inference version of MJFA. Specifically, we introduce the SimKD algorithm [17] as a basis to ensure the discriminative power of the distilled knowledge. In addition, we propose a tri-path collaborative alignment strategy that effectively retains the domain invariance of the distilled knowledge by comprehensively constraining cross-domain consistency between the teacher and student models as well as within the student model itself. Extensive experiments conducted in two transductive scenarios and one inductive scenario on the SAMPLE dataset [5] validate the effectiveness of the proposed MFJA framework and DCKD algorithm.

The core contributions are summarized as follows.

1) The SFFN and PA-GAT effectively integrate domain knowledge from both the modalities into feature extraction, while the GFFM enhances fusion effectiveness via adaptive recalibration. These components work in a complementary manner, jointly promoting more accurate and domain-invariant feature representations.
2) The proposed PN-WGRL is capable of guiding the model to pay greater attention to hard-to-align samples during global domain alignment, thereby facilitating the LMMD loss in achieving more precise local domain alignment.
3) The tri-path collaborative alignment strategy within DCKD effectively ensures the domain invariance of the distilled knowledge by enforcing more comprehensive cross-domain constraints, significantly enhancing the cross-domain recognition performance of the resulting lightweight model.

The rest of this article is structured as follows. Section II provides a brief review of related works. The proposed MFJA is described in detail in Section III. Section IV presents the proposed DCKD algorithm. The experimental results and their corresponding analyses are provided in Section V. In conclusion, Section VI summarizes the findings of this article.

II. Related Works

A. Domain Adaptation and Related Algorithms

Domain adaptation methods have attracted increasing attention in recent years due to their ability to transfer knowledge from a source domain to a target domain with different data distributions, thereby extending the applicability of machine learning models. In the field of SAR ATR, this capability is particularly valuable for bridging the distributional gap between synthetic and measured images.

Recently, various domain adaptation methods have been proposed. He et al. [6] designed a task-driven domain adaptation method that uses the maximum mean discrepancy loss [18] to address distributional shifts induced by varying imaging depression angles. Du et al. [7] integrated image reconstruction with domain adversarial training, proposing an adversarial coding network. However, this method requires access to a small number of labeled measured images. Lyu et al. [10] presented an unsupervised method that combines the maximum mean discrepancy loss with domain adversarial training. Despite its effectiveness at aligning global distributions, it overlooks potential subdomain shifts. In a subsequent study, Lv et al. [9] introduced a domain adaptation algorithm that integrates dual-branch reconstruction with subdomain alignment. Nevertheless, it relies on a few labeled measured samples during the initial training stage to facilitate more precise subdomain alignment. Sun et al. [11] proposed a gradual domain adaptation recognition framework that incorporates pseudolabeled measured images into the supervised classification process to refine local distributions. However, their method does not incorporate SAR domain knowledge. Shi et al. [12] further introduced the ASC model [19], specifically designed for SAR imagery, to generate more accurate pseudolabels by leveraging domain knowledge. Nevertheless, the extraction of ASCs is computationally intensive. Moreover, the domain knowledge is not explicitly incorporated into the feature extraction process. Our previous study [13] proposed a VSFA. This method integrates visual features extracted by the CNN with two different scattering topological features extracted by the GNN, where the corresponding scattering topological graphs are constructed based on the ASC model and the SAR-SIFT algorithm [20], thereby explicitly incorporating SAR domain knowledge into the feature extraction process. The designed two-stage loss enables progressive domain alignment from global to local levels. Despite these efforts, this method still fails to fully exploit domain knowledge and neglects the varying alignment difficulties across different samples.

B. Multimodal Representation Learning and Related Algorithms

Multimodal representation learning [21] refers to a class of methods that construct richer and more discriminative

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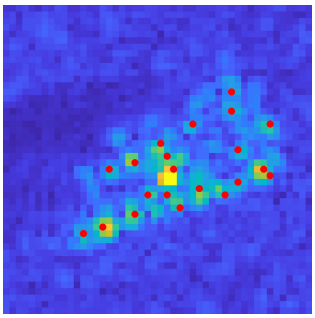


Fig. 3. Scattering topological points extracted by the SAR-SIFT algorithm.

representations by fusing information across diverse modalities (including, but not limited to, visible images [22], infrared images [23], and SAR images [24]). The utilization of complementary strengths among modalities improves the accuracy and robustness of the model in complex tasks. Recently, multimodal representation learning has seen rapid development within the field of SAR ATR, extending its scope from the fusion of data acquired by different sensors to the fusion of heterogeneous representations derived from the same data.

As illustrated in Fig. 3, the distribution of scattering topological points can be used to characterize the whole topological structure of the target to a certain extent. As a result, the scattering topological modality has become another widely adopted modality in addition to visual modality. Qin et al. [25] designed a scattering attribute embedded network for few-shot SAR image classification, which comprised an attribute induced global semantic information extraction branch and a graph-based structure representation learning branch. A scattering topology network was proposed by Kang et al. [26] for aircraft classification. Their design features a scattering topology module that learns the spatial relationships among discrete scattering points, alongside a context attention excitation mechanism that emphasizes critical global vision information. Zhang et al. [27] presented a multiscale global scattering feature association network. It used multikernel convolutional layers for visual feature extraction and combined a scattering center feature association module and a multiscale feature enhancement module for scattering topological feature extraction. Wen et al. [28] introduced the phase history modality of SAR images in addition to visual modality and scattering topological modality, proposing a multimodal discriminative feature learning framework. Our previous study [13] combined ASC-based topological graphs and SAR-SIFT-based topological graphs and demonstrated that their complementary information can enhance recognition performance. However, the extraction of ASCs is computationally expensive, which limits the efficiency of ASC-based multimodal representation learning algorithms. Moreover, keypoint-based algorithms often fail to fully leverage the rich information embedded in keypoints, including their spatial distribution and local scattering characteristics.

C. KD and Related Algorithm

The core idea of KD aims to instruct the student model to emulate the output of the teacher model, thus allowing the student model to learn knowledge similar to that of the teacher model while maintaining a small model size. Vanilla KD operates within single-modality and single-domain contexts, focusing only on the discriminative ability of the distilled knowledge. It has been extensively studied in SAR ATR [29], [30]. When extending to cross-modal KD, current approaches generally focus on transferring knowledge between optical and SAR data [31], [32]. Their scenarios do not involve two different domains (e.g., day and night, and synthetic and measured), thereby still focusing solely on the discriminative ability of the distilled knowledge. Unlike the above algorithms, the primary challenge in our scenario lies in ensuring that the knowledge distilled from a multimodal model into a lightweight unimodal model retains both discriminative power and domain invariance across synthetic and measured SAR images. To the best of our knowledge, this dual focus has not been adequately explored in prior SAR ATR literature.

In the broader field of computer vision, some studies have explored cross-modal KD with domain invariance. Li et al. [33] proposed a cross-modal KD algorithm for 3-D semantic segmentation that operates across both the modalities and domains. Despite this advancement, their approach limits the teacher models to only accepting inputs from the target domain, thus failing to explicitly align source domain teacher features with target domain student features. In addition, their approach neglects the domain alignment within the student models, thereby limiting its effectiveness in cross-domain generalization.

III. Multimodal Feature Fusion and Global–Local Joint Alignment

A. Problem Definition

In the context of unsupervised domain adaptation tasks, there are two similar yet distinct domains: a source domain $\mathcal{D}^s = \{(\mathbf{X}_i^s, \mathbf{y}_i^s)\}_{i=1}^{n^s}$ with n^s labeled samples and a target domain $\mathcal{D}^t = \{\mathbf{X}_j^t\}_{j=1}^{n^t}$ with n^t unlabeled samples, where the superscripts $(\cdot)^s$ and $(\cdot)^t$ distinguish between the source and target domains. The symbol $\mathbf{y}_i^s \in \mathbb{R}^L$ represents a one-hot vector that indicates the ground-truth class label of the ith source sample $\mathbf{X}_i^s$, where L denotes the number of classes. The objective of unsupervised domain adaptation is to diminish the distributional differences by leveraging labeled source domain data and unlabeled target domain data, thereby assisting the model in minimizing the prediction error $\sum_{j=1}^{n^t} \|\bar{\mathbf{y}}_j^t - \mathbf{y}_j^t\|$ in the target domain. The symbols $\bar{\mathbf{y}}_j^t$ and $\mathbf{y}_j^t$ denote the predicted and true class labels of the jth target sample $\mathbf{X}_j^t$, respectively. In particular, the latter is unavailable during training.

In addition, each sample has a ground-truth domain label $\mathbf{d} \in \mathbb{R}^2$. For samples from the source domain, the domain label $\mathbf{d}$ is set to $(1, 0)$, whereas for those from the target domain, it is defined as $(0, 1)$. And the symbol $\bar{\mathbf{d}} = (\bar{d}_s, \bar{d}_t)$ denotes the predicted domain label, where $\bar{d}_s$ and $\bar{d}_t$ represent the confidences of a sample being predicted to originate from the source domain and the target domain, respectively.

B. Overall Architecture

The architecture of the proposed MFJA is depicted in Fig. 4. As can be seen, it comprises four distinct modules: a

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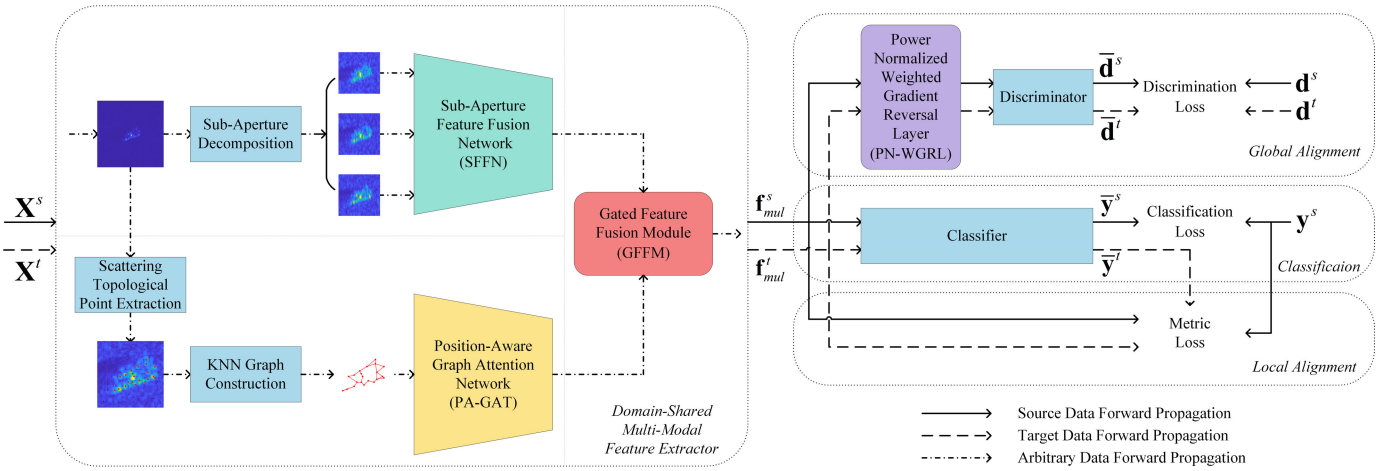


Fig. 4. Overall architecture of MFJA, which comprises four components.

multimodal feature extractor, a classification module, a global alignment module, and a local alignment module.

The multimodal feature extractor is designed to extract the multimodal feature $\mathbf{f}_{\text{mul}}$ from an arbitrary complex-valued full-aperture SAR image $\mathbf{X}$. It comprises three subcomponents: 1) a visual feature extraction branch, which includes subaperture decomposition followed by an SFFN; 2) a scattering topological feature extraction branch, which involves scattering topological point extraction, scattering topological graph construction, and a PA-GAT; and 3) a GFFM that adaptively fuses the outputs from the above two branches.

The classification module consists of a classifier and a classification loss, which quantifies the differences between predicted probabilities and true labels. Note that the classification loss is computed only on source domain samples.

The global alignment module uses adversarial learning to align distributions at the domain level. It includes a PN-WGRL, a discriminator, and a discrimination loss.

The local alignment module focuses on aligning distributions at the class level through a divergence-based approach and is implemented solely via a metric loss.

In Sections III-C–III-H, we will begin by introducing the three subcomponents of the multimodal feature extractor. Subsequently, as the global alignment module is designed to assist the local alignment module, the local alignment module will be introduced first, followed by the global alignment module, and finally, the classification module.

C. Visual Feature Extraction

Most algorithms, including our previously proposed VSFA, typically overlook the incorporation of domain knowledge in visual feature extraction. In this article, we introduce the subaperture decomposition method [14] and design an SFFN to fully exploit the subaperture information of SAR images, thereby enhancing the discrimination ability of visual features. The specifics are presented as follows.

1) Subaperture Decomposition: The physical size of antennas on aircraft and satellites imposes a fundamental limit on the achievable azimuth resolution in radar imaging. To overcome this limitation, SAR synthesizes a long virtual antenna by leveraging the Doppler effect of echoes as the radar platform moves, thereby achieving high-resolution azimuth imaging. However, a tradeoff arises: the resulting full-aperture image cannot resolve the variations in target backscattering at individual azimuths. To address this issue, the subaperture decomposition method has been developed, which enables the decomposition of the full-aperture image into multiple subaperture images, each corresponding to a narrower range of azimuths.

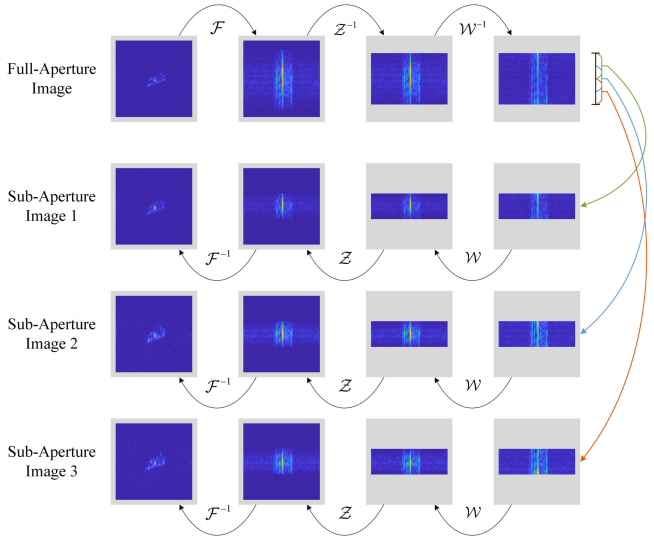


Fig. 5. Flowchart of subaperture decomposition (with $c = 3$). Different colors represent different spectrum ranges.

The process of subaperture decomposition is depicted in Fig. 5, where $\mathcal{F}$, $\mathcal{Z}$, and $\mathcal{W}$ represent the Fourier transform, zero padding, and windowing, respectively. In addition, $(\cdot)^{-1}$ represents the inverse operation of $(\cdot)$. First, the 1-D Fourier transform is performed along azimuth to convert the image into the range–Doppler domain. Second, the zero elements added to smooth the image are removed. Third, the spectrum is multiplied by the inverse of the window function to recover the uniformly distributed backscattering power. Specifically, referring to the imaging parameters of the SAMPLE dataset [5], the

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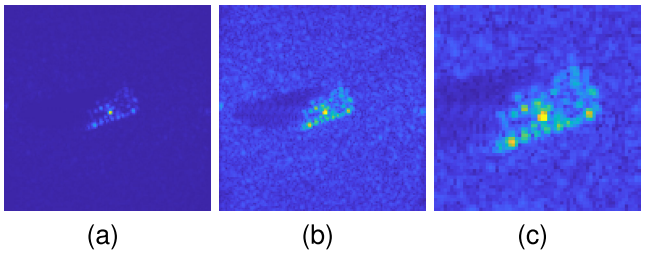


Fig. 6. Visualization of quarter power magnitude normalization and cropping. (a) Original image. (b) Normalized image. (c) Cropped image.

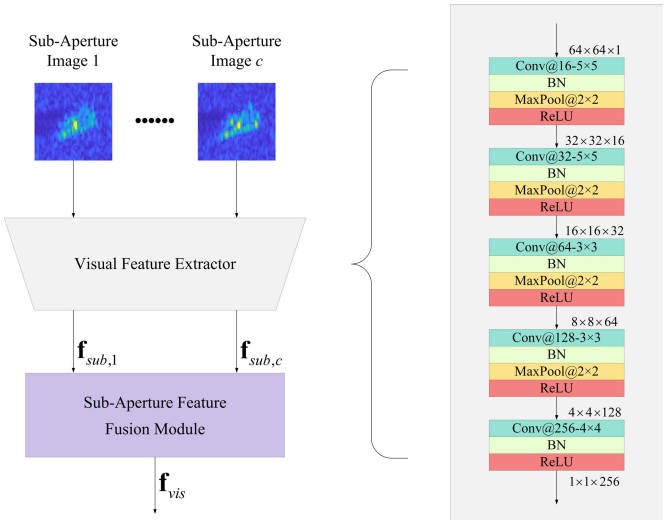


Fig. 7. Overall architecture of the designed SFFN. Conv@16-5 × 5 indicates that 16 convolution kernels of size 5 × 5 are used to perform convolution on the input feature map.

Taylor window with −35-dB sidelobe suppression is adopted. Fourth, the spectrum is divided equally spaced into c partially overlapping subbands along azimuth, each containing half of the whole spectrum. Finally, the inverse operations of the first three operations described above are performed to generate subaperture images, including multiplying by the Taylor window, padding with zero elements, and applying the 1-D inverse Fourier transform.

Unfortunately, the generated images exhibit low contrast, which presents a challenge for subsequent feature extraction. Following the authors of the SAMPLE dataset, the quarter power magnitude normalization is used to address this issue. The original and normalized images are illustrated in Fig. 6(a) and (b), respectively. In addition, the normalized image is centrally cropped into a size of 64 × 64 pixels to reduce the computational complexity before being fed into the SFFN, as shown in Fig. 6(c).

2) Subaperture Feature Fusion Network: After the full-aperture image is decomposed into subaperture images, the SFFN is designed to extract visual features from subaperture images and then fuse them to obtain a more discriminative visual feature $\mathbf{f}_{\text{vis}}$. As illustrated in Fig. 7, the SFFN comprises two components: an interaperture shared visual feature extractor and a subaperture feature fusion module. The former is a simple five-layer network comprising the convolutional layer, the BN layer [34], the max pooling layer, and the ReLU activation function [35]. Three distinct

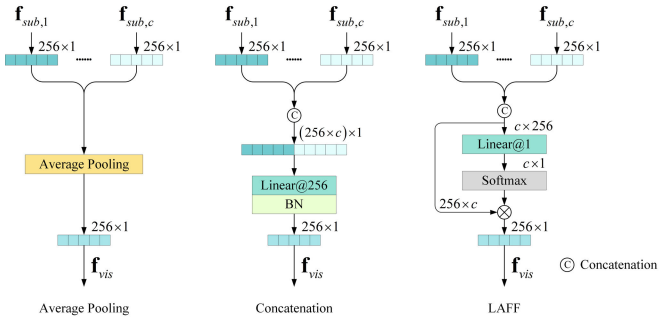


Fig. 8. Three distinct methods for subaperture feature fusion. Linear@256 indicates that a linear layer is applied to transform the input feature to 256-D.

methods are considered for the latter: average pooling, concatenation, and LAFF [36]. The details of these three methods are presented in Fig. 8. The average pooling method averages all the subaperture features along the aperture dimension. The concatenation method involves concatenating the subaperture features and using a linear layer and a BN layer for information compression. The LAFF method calculates the normalized weights of the subaperture features through a linear layer and the softmax function, followed by the weighted summation of the subaperture features. In light of the computational efficiency and recognition performance, the average pooling method is chosen for subaperture feature fusion. The detailed experiments are shown in Section V-E.

D. Scattering Topological Feature Extraction

As established in Section II, scattering topological information benefits SAR target recognition. The general extraction pipeline includes extracting scattering topological points, constructing graphs, and designing a GNN for feature learning. For scattering topological point extraction, the ASC model suffers from high computational complexity, limiting its practical application. Therefore, we adopt more efficient keypoint detection algorithms. Unfortunately, most existing keypoint-based methods only use either spatial positions or descriptors, failing to exploit their full potential. This shortcoming drives us to propose a PA-GAT that effectively integrates both types of information to enhance the discrimination ability of scattering topological features. The specifics are described below.

1) Scattering Topological Point Extraction: Benefiting from the proposed SAR-Harris detector and ratio descriptor, the SAR-SIFT algorithm effectively mitigates the adverse effects of multiplicative speckle noise on both keypoint detection and descriptor generation, thereby adapting the classical SIFT algorithm [37] for effective application in SAR image processing. This work follows the same SAR-SIFT-based procedure for extracting scattering topological points as described in our previously proposed VSFA framework. First, the SAR-Harris detector is applied to extract candidate keypoints. Given that the number of extracted keypoints is relatively limited, the clustering algorithm [38] is further used to increase their quantity, thereby enabling a more comprehensive description of the target structure. Subsequently, a two-step filtering algorithm is performed to eliminate the inaccurate keypoints on the background region, resulting in the final set of scat-

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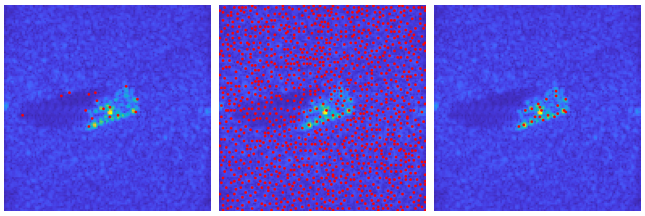


Fig. 9. Visualization of keypoints. (a) Keypoints extracted by the SAR-Harris detector. (b) Keypoints extracted by the clustering algorithm. (c) Keypoints after the two-step filtering.

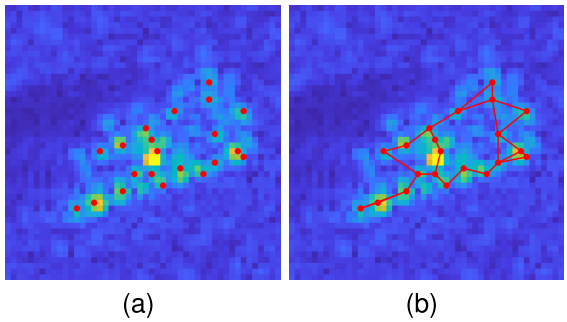


Fig. 10. Visualization of scattering topological graph. (a) Scattering topological points. (b) Corresponding topological graph.

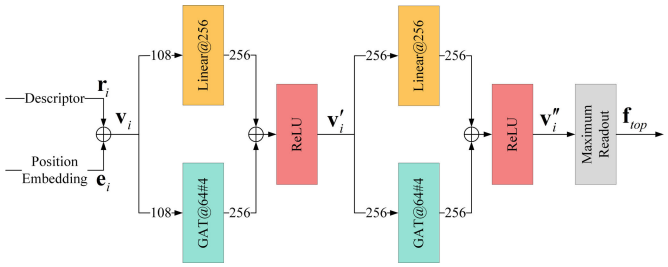


Fig. 11. Overall architecture of the designed PA-GAT. GAT@64#4 indicates the utilization of a four-head graph attention convolution layer, with each attention head projecting features into 64-D.

tering topological points. Finally, the 108-D ratio descriptor is generated for each point to characterize its local scattering properties in gradient form.

Note that before extracting scattering topological points, the quarter power magnitude normalization is used to enhance the contrast of the full-aperture SAR image. In addition, the number of extracted scattering topological points may vary in different images. Further implementation details can be found in [13] and [20]. Fig. 9 visualizes the scattering topological points before and after the two-step filtering, demonstrating their capability in capturing the target's topological structure.

2) Scattering Topological Graph Construction: Without loss of generality, assume that m scattering topological points are extracted from an SAR image. Consequently, the target can be effectively represented as a set of points $\mathcal{V} = \{v_i\}_{i=1}^{m}$, where v_i indicates the ith point and is characterized by three features: the ratio descriptor $\mathbf{r}_i$, the range coordinate p_i, and the azimuth coordinate q_i. Given the non-Euclidean properties inherent in the set of scattering topological points, a graph structure is introduced to model its scattering topology. Conventionally, a graph $\mathcal{G}$ is defined as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ and $\mathcal{E}$ represent a set of nodes and a set of edges, respectively. Therefore, to generate a scattering topological graph, it is sufficient to define the edges between these points. Specifically, we initially calculate the Euclidean distance between each pair of scattering topological points by their coordinates (p, q). Subsequently, unidirectional edges are established from each scattering topological point to its three nearest points. Consequently, the directed unweighted graph is obtained, encapsulating both the local backscattering characteristics of the target and its global spatial structure. Fig. 10 illustrates a set of scattering topological points and its corresponding topological graph.

3) PA-GAT: GNNs are capable of learning spatial topology during node information aggregation, leading to their widespread applications in graph learning. To accurately capture the scattering structure of the target by leveraging both the spatial position and descriptor of each node, we propose a novel PA-GAT, which extends the widely used GAT framework [39].

Fig. 11 illustrates the overall architecture of the proposed PA-GAT. It is observed that the initial node feature $\mathbf{v}_i$ of the ith node is a sum of its descriptor $\mathbf{r}_i$ and its position embedding $\mathbf{e}_i$. Indeed, a straightforward method would be to concatenate the descriptor $\mathbf{r}_i$ and the coordinate (p_i, q_i). However, the imbalanced dimension ratio between the 108-D descriptor and the 2-D position coordinate poses a challenge for effectively capturing location information during feature extraction. Therefore, inspired by transformer architectures, the sinusoidal position embedding method [40] is introduced to project position coordinates into a high-dimensional space that aligns with the dimensionality of the descriptors, facilitating a more robust fusion by elementwise summation. Given that the position of the ith node is described by both an range coordinate p_i and a azimuth coordinate q_i, these two coordinates are first projected into a 54-D space. Their high-dimensional representations are then concatenated to form the position embedding $\mathbf{e}_i$. Finally, the position embedding $\mathbf{e}_i$ and the descriptor $\mathbf{r}_i$ are summed to generate the initial node feature $\mathbf{v}_i$. In particular, the L2 normalized descriptor is renormalized to $[-1, 1]$ by min–max normalization before summing to align with the value range of position embedding. The whole process can be described as

$$\mathrm{PE}(\mathrm{pos})_k = \begin{cases} \sin\left(\mathrm{pos}/10000^{k/54}\right), & \text{if } k \text{ even} \\ \cos\left(\mathrm{pos}/10000^{(k-1)/54}\right), & \text{if } k \text{ odd} \end{cases} \tag{1}$$

$$\mathbf{e}_i = \mathrm{PE}(p_i) || \mathrm{PE}(q_i) \tag{2}$$

$$\mathbf{v}_i = \frac{1}{2}(\mathbf{r}_i + \mathbf{e}_i) \tag{3}$$

where $\mathrm{PE}(\mathrm{pos})_k$ represents the kth element of the position embedding corresponding to the position coordinate pos, and $||$ denotes the concatenation operation.

In addition, to mitigate the common phenomenon of oversmoothing in graph representation learning, we design an additional linear branch in the network structure, which works in conjunction with the graph attention convolution branch. The linear branch aims to perform individual transformations on each node, thereby enhancing the feature extraction capability while preserving the diversity of features among nodes.

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Without loss of generality, taking the first layer as an example, the aggregation process for the ith node is shown as follows:

$$\mathbf{v}_i' = \max(\text{Linear}(\mathbf{v}_i) + \text{GAT}(\mathbf{v}_i), 0) \tag{4}$$

$$\text{Linear}(\mathbf{v}_i) = \mathbf{W}_1\mathbf{v}_i + \mathbf{b}_1 \tag{5}$$

$$\text{GAT}(\mathbf{v}_i) = \|_{h=1}^{H} \sum_{j\in\mathcal{N}_i} \alpha_{ij}^h \mathbf{W}_G^h \mathbf{v}_j \tag{6}$$

$$\alpha_{ij}^h = \frac{\exp\left(\text{LeakyReLU}\left(\left(\mathbf{a}^h\right)^{\mathrm{T}}\left[\mathbf{W}_G^h\mathbf{v}_i \| \mathbf{W}_G^h\mathbf{v}_j\right]\right)\right)}{\sum_{l\in\mathcal{N}_i}\exp\left(\text{LeakyReLU}\left(\left(\mathbf{a}^h\right)^{\mathrm{T}}\left[\mathbf{W}_G^h\mathbf{v}_i \| \mathbf{W}_G^h\mathbf{v}_l\right]\right)\right)} \tag{7}$$

where $\mathbf{v}_i'$ is the aggregated node feature. $\mathbf{W}_1$ and $\mathbf{b}_1$ are the weight matrix and bias vector of the linear layer, respectively. $\mathbf{W}_G$ and $\mathbf{a}$ are the learnable parameters used to transform features and compute attention coefficients in the graph attention layer. H indicates the number of attention heads. $\mathcal{N}_i$ denotes the set of nodes connected to the ith node. $(\cdot)^{\mathrm{T}}$ represents the transposition operation. The Leaky ReLU activation function [35] is used to ensure that gradients are propagated back even when attention coefficients are negative, prior to softmax normalization. Specifically, the four-head attention mechanism is used in PA-GAT, with each attention head projecting node features into 64-D. Following two layers of feature extraction, the maximum readout function is applied to perform max pooling along the node dimension to aggregate information from all the nodes, thereby generating a scattering topological feature $\mathbf{f}_{\text{top}}$ containing the scattering characteristics of the target and its spatial topology.

E. Gated Feature Fusion Module

As visual features and scattering topological features originate from different modalities, they often contain both complementary and conflicting information. Consequently, simple concatenation or average is generally not the optimal fusion method. Inspired by the gating mechanism [41], a GFFM is developed for adaptive fusion.

The overall architecture of the GFFM is presented in Fig. 12. First, an intermodal shared linear layer is applied to learn a weight vector for each modal feature. The weight vector modulates each element of the corresponding modal feature, thereby suppressing conflicting information while mining complementary information. Subsequently, the modulated features $\mathbf{f}_{\text{mvis}}$ and $\mathbf{f}_{\text{mtop}}$ are concatenated and then fed into a subsequent linear layer for dimension compression, thereby removing redundant information. Finally, the ReLU activation function is used to improve the nonlinear mapping ability. The fusion process can be mathematically represented as

$$\mathbf{f}_{\text{mvis}} = \mathbf{f}_{\text{vis}} \odot (\mathbf{W}_2\mathbf{f}_{\text{vis}} + \mathbf{b}_2) \tag{8}$$

$$\mathbf{f}_{\text{mtop}} = \mathbf{f}_{\text{top}} \odot \left(\mathbf{W}_2\mathbf{f}_{\text{top}} + \mathbf{b}_2\right) \tag{9}$$

$$\mathbf{f}_{\text{mul}} = \max\left(\mathbf{W}_3\left[\mathbf{f}_{\text{mvis}} \| \mathbf{f}_{\text{mtop}}\right] + \mathbf{b}_3, 0\right) \tag{10}$$

where $\mathbf{f}_{\text{mul}}$ represents the multimodal fusion feature and $\odot$ denotes the Hadamard product. $\mathbf{W}_2$, $\mathbf{b}_2$, $\mathbf{W}_3$, and $\mathbf{b}_3$ are the learnable parameters of the two linear layers.

With the synergy of the three components mentioned above, the designed multimodal feature extractor enables the effective extraction of a multimodal representation $\mathbf{f}_{\text{mul}}$ from an arbitrary complex-valued full-aperture SAR image $\mathbf{X}$, thereby providing a discriminative feature space for domain alignment. Subsequently, both the global domain-level and local class-level distributions are explicitly aligned to bridge the distributional gap between the source and target domains, facilitating better generalization of the classifier trained on the source domain to the target domain.

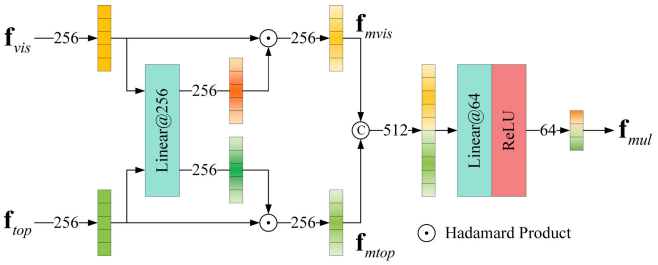


Fig. 12. Overall architecture of the designed GFFM.

F. Local Alignment

Consistent with VSFA, the widely used LMMD loss is used for the purpose of aligning the local distributions at the class level. The LMMD loss facilitates the class-by-class alignment by assigning the unlabeled target domain samples to all the classes with different probabilities. It is defined as

$$\begin{aligned}
&\mathcal{L}_{\text{lmmd}}\left(\mathbf{f}_{\text{mul}}^{s}, \mathbf{f}_{\text{mul}}^{t}\right) \\
&= \frac{1}{L}\sum_{l=1}^{L} \| \sum_{i=1}^{n^s} \omega_i^{sl}\phi\left(\mathbf{f}_{\text{mul},i}^{s}\right) - \sum_{j=1}^{n^t} \omega_j^{tl}\phi\left(\mathbf{f}_{\text{mul},j}^{t}\right)\|_{\mathcal{H}}^2 \\
&= \frac{1}{L}\sum_{l=1}^{L}\left[\sum_{i=1}^{n^s}\sum_{j=1}^{n^s}\omega_i^{sl}\omega_j^{sl}k\left(\mathbf{f}_{\text{mul},i}^{s}, \mathbf{f}_{\text{mul},j}^{s}\right)\right. \\
&\qquad + \sum_{i=1}^{n^t}\sum_{j=1}^{n^t}\omega_i^{tl}\omega_j^{tl}k\left(\mathbf{f}_{\text{mul},i}^{t}, \mathbf{f}_{\text{mul},j}^{t}\right) \\
&\qquad \left. -2\sum_{i=1}^{n^s}\sum_{j=1}^{n^t}\omega_i^{sl}\omega_j^{tl}k\left(\mathbf{f}_{\text{mul},i}^{s}, \mathbf{f}_{\text{mul},j}^{t}\right)\right]
\end{aligned} \tag{11}$$

where L refers to the number of classes. $\mathcal{H}$ denotes the RKHS generated by a kernel function k. The symbol $\phi(\cdot)$ represents the feature mapping function that maps multimodal features to the RKHS and is defined as

$$k\left(\mathbf{f}_{\text{mul},i}^{s}, \mathbf{f}_{\text{mul},j}^{t}\right) = \langle\phi\left(\mathbf{f}_{\text{mul},i}^{s}\right), \phi\left(\mathbf{f}_{\text{mul},j}^{t}\right)\rangle \tag{12}$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product of vectors. Referring to [15], a Gaussian kernel is used here, with its bandwidth set to the median pairwise squared distances calculated from both the source and target domains. ω_i^{sl} and ω_i^{tl} denote the weights of $\mathbf{f}_{\text{mul},i}^{s}$ and $\mathbf{f}_{\text{mul},i}^{t}$ in the computation of the mean embedding for class l. Their calculation processes can be expressed as

$$\omega_i^{sl} = \frac{y_i^{sl}}{\sum_{j,\left(\mathbf{x}_j^s, \mathbf{y}_j^s\right)\in\mathcal{D}^s} y_j^{sl}} \tag{13}$$

$$\omega_i^{tl} = \frac{\bar{y}_i^{tl}}{\sum_{j,\left(\mathbf{x}_j^t, \bar{y}_j^t\right)\in\mathcal{D}^t} \bar{y}_j^{tl}} \tag{14}$$

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where y_i^{sl} denotes the lth element of the ground-truth class label $\mathbf{y}_i^s$ associated with the ith source domain sample $\mathbf{X}_i^s$, and $\bar{y}_i^{tl}$ denotes the lth element of the predicted class label $\bar{y}_i^t$ associated with the ith target domain sample $\mathbf{X}_i^t$.

It is observed that the LMMD loss essentially quantifies, on a per-class basis, the discrepancy between the mean embeddings of the source and target domains within the RKHS. The mean embedding for each class is computed as the normalized weighted sum of the embeddings of the samples within that class. Each sample's contribution to the mean embedding depends on its predicted probability of belonging to that class, as well as the probabilities of all other samples in the same class, as shown in (13) and (14). In the source domain, where class labels are known, each sample contributes exclusively to the mean embedding of its true class. However, in the target domain, where labels are not available during training, the LMMD loss assigns each sample to all the categories based on its predicted confidences, allowing it to contribute to the mean embedding computation for multiple categories with different weights. It is evident that a misclassified sample will have a greater impact on the incorrect class than on its true class, leading to biased local domain alignment.

G. Global Alignment

To alleviate the harmful effects of misclassified samples on local domain alignment, VSFA proposed a two-stage loss function. During the early stage of training, the improved CORAL loss is used to reduce global distributional differences. Subsequently, in the mid and late stages of training, the LMMD loss is applied to refine the local distributions based on the initial global alignment. Despite some progress, the improved CORAL loss treats all the samples equally, lacking the ability to mine hard-to-align samples, which are crucial for improving local domain alignment.

Recently, Shi et al. [12] propose a WGRL, building up on the original GRL [42], which is inspired by the generative adversarial network [43]. The core objective of GRL is to extract domain-invariant features by confusing the discriminator through gradient reversal during backward propagation. As shown in Fig. 4, the GRL is generally inserted between the feature extractor and the discriminator. Without loss of generality, for the ith multimodal feature $\mathbf{f}_{\text{mul},i}$, the GRL acts as an identity mapping in the forward propagation

$$\text{GRL}\left(\mathbf{f}_{\text{mul},i}\right) = \mathbf{f}_{\text{mul},i} \tag{15}$$

while reversing gradients during backward propagation

$$\frac{d\,\text{GRL}\left(\mathbf{f}_{\text{mul},i}\right)}{d\mathbf{f}_{\text{mul},i}} = -\lambda\mathbf{I} \tag{16}$$

where $\mathbf{I}$ denotes an identity matrix and λ represents the weight coefficient. This mechanism effectively unifies two conflicting objectives: optimizing the discriminator to distinguish domains, while forcing the feature extractor to confuse it by reversing gradients.

However, the GRL also ignores the fact that the alignment difficulty varies across samples. To tackle this, the WGRL assigns a sample-specific weight to the ith sample $\mathbf{X}_i$, based on its alignment difficulty, measured by the information entropy of the predicted domain label $\bar{\mathbf{d}}_i$. This allows the model to assign higher importance to samples that are more difficult to align, facilitating a more precise domain alignment. Similar to the GRL, the WGRL preserves the identity mapping during forward propagation, but scales gradients during backward propagation

$$\frac{d\,\text{WGRL}\left(\mathbf{f}_{\text{mul},i}\right)}{d\mathbf{f}_{\text{mul},i}} = -\lambda\mathbf{I}\cdot(1-u_i) \tag{17}$$

where u_i represents the information entropy of the predicted probability $\bar{\mathbf{d}}_i$ and is defined as

$$u_i = -\langle\bar{\mathbf{d}}_i, \log\left(\bar{\mathbf{d}}_i\right)\rangle. \tag{18}$$

A higher entropy value reflects greater uncertainty in the discriminator's domain prediction, suggesting that the features extracted from a sample are already domain-invariant. As a result, the gradient associated with this sample is scaled down during backpropagation to reduce its influence on further feature learning. Conversely, a sample with lower entropy, whose features remain domain-specific, retains a larger gradient to encourage better alignment.

However, our investigation into the patterns of weight $1-u$ variations under different probability distributions $\bar{\mathbf{d}}$ reveals that this weighting approach excessively reduces the gradients of certain hard-to-align samples. As shown in Fig. 13(a), the slopes at both the ends of the weighting curve are considerable, which causes the network to focus predominantly on samples that are extremely difficult to confuse. As a result, relatively hard-to-align samples receive insufficient attention, despite their importance for effective domain adaptation. For instance, a sample with a discrimination probability of (0.9, 0.1) or (0.1, 0.9) is actually a hard-to-align sample, but its weight is excessively reduced to near 0.5.

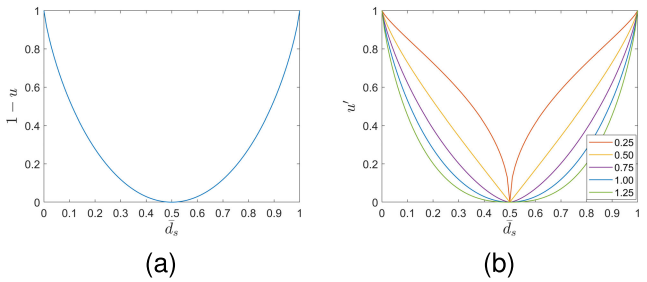


Fig. 13. Variation in weight across different probability distributions. (a) Original weight curve. (b) Weight curves with different powers.

To address this problem, we introduce the power transformation to remap the weights, which is defined as

$$u_i' = (1-u_i)^{\beta} \tag{19}$$

where u_i' represents the transformed weight and β is the power. Fig. 13(b) depicts the variation curves of the weights with different powers. The results indicate that a decrease in power corresponds to a reduction in the slope of the weight curve at its ends, thus boosting the weights of the relatively hard-to-align samples.

Unfortunately, the power transformation also amplifies the weights of relatively easy-to-align samples. To further

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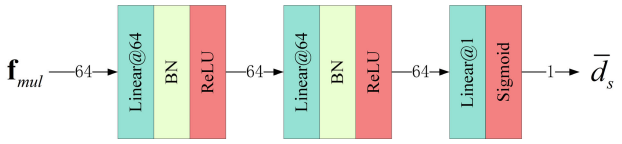


Fig. 14. Overall architecture of the designed discriminator.

regulate the weights of samples with varying alignment difficulties, we introduce the temperature-scaled softmax normalization [16]. Specifically, the softmax normalization is performed on the power transformed weights u' among all the samples of a batch $\mathcal{B}$, simultaneously controlling the smoothness of the output through the temperature coefficient τ. It is defined as

$$u_i'' = \frac{\exp\left(u_i'/\tau\right)}{\sum_{j\in\mathcal{B}}\exp\left(u_j'/\tau\right)} \tag{20}$$

where u'' is the normalized weight. In conclusion, we propose a novel PN-WGRL based on the GRL and the WGRL. It is still equivalent to the identity transformation during forward propagation, but further scales gradients during backward propagation

$$\frac{d\,\text{PN-WGRL}\left(\mathbf{f}_{\text{mul},i}\right)}{d\mathbf{f}_{\text{mul},i}} = -\lambda\mathbf{I}\cdot u_i''. \tag{21}$$

The designed discriminator is a simple three-layer network, whose architecture is shown in Fig. 14. In fact, the output of the discriminator is the probability $\bar{d}_s$ that the sample $\mathbf{X}$ originates from the source domain. In consideration of the fact that there are only two domains, the probability $\bar{d}_t$ that the sample $\mathbf{X}$ originates from the target domain is denoted as $1-\bar{d}_s$. And the discrimination loss, which is essentially a binary cross-entropy loss, is defined as

$$\mathcal{L}_{\text{dis}} = -\frac{1}{2}\left(\frac{1}{n^s}\sum_{i=1}^{n^s}\langle\mathbf{d}_i^s, \log\left(\bar{\mathbf{d}}_i^s\right)\rangle + \frac{1}{n^t}\sum_{j=1}^{n^t}\langle\mathbf{d}_j^t, \log\left(\bar{\mathbf{d}}_j^t\right)\rangle\right). \tag{22}$$

H. Classification

The classification module comprises a classifier and a classification loss. The former is a linear layer with the softmax function. The latter is a general multiclass cross-entropy loss and is defined as

$$\mathcal{L}_{\text{cls}} = -\frac{1}{n^s}\sum_{i=1}^{n^s}\langle\mathbf{y}_i^s, \log\left(\bar{\mathbf{y}}_i^s\right)\rangle \tag{23}$$

where $\mathbf{y}_i^s$ and $\bar{\mathbf{y}}_i^s$ denote the ground truth and predicted class labels of the ith source domain sample $\mathbf{X}_i^s$, respectively.

To summarize, the overall loss function can be formulated as

$$\mathcal{L}_{\text{mul}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{dis}} + \lambda_1\cdot\mathcal{L}_{\text{lmmd}} \tag{24}$$

where λ_1 is the weight coefficient.

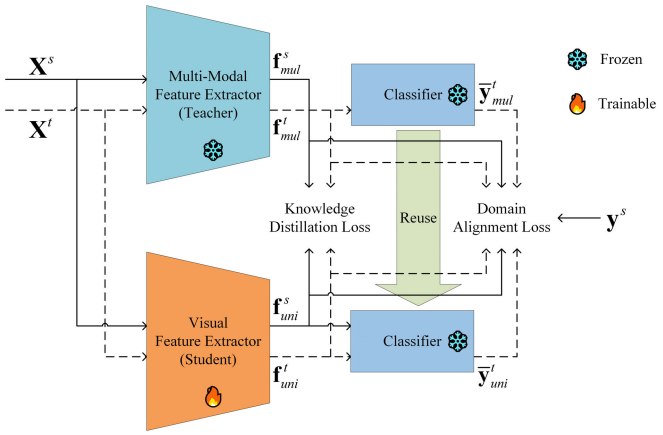


Fig. 15. Overall architecture of the proposed DCKD algorithm.

IV. Domain-Invariant Cross-Modal KD

The incorporation of SAR domain knowledge, including subaperture and scattering topology information, significantly enhances MFJA's performance. However, this comes with increased inference time, somewhat limiting its practical applicability. To enable efficient large-scale inference on unseen samples in inductive scenarios, we propose a DCKD algorithm. The purpose of DCKD is to distill both discriminative and domain-invariant knowledge from the multimodal teacher model into the visual student model based on full-aperture images, thereby eliminating the need for subaperture decomposition and scattering topological graph construction during inference while maintaining cross-domain recognition performance. The overall architecture of the proposed DCKD algorithm is illustrated in Fig. 15, where the subscripts $(\cdot)_{\text{mul}}$ and $(\cdot)_{\text{uni}}$ are intended to distinguish between multimodal and visual unimodal. Notably, only the visual feature extractor needs to be trained from scratch; its detailed architecture is presented in Fig. 7.

Specifically, the SimKD algorithm [17] is adopted into the cross-modal KD scenario as the basis for distillation. It argues that the classifier of the teacher model is sufficiently discriminative. Therefore, it directly reuses the classifier for student inference and trains a student feature extractor through KD based on the MSE loss. And the MSE loss is defined as

$$\mathcal{L}_{\text{kd}} = \frac{1}{2}\left(\frac{1}{n^s}\sum_{i=1}^{n^s}\left\|\mathbf{f}^s_{\text{mul}} - \mathbf{f}^s_{\text{uni}}\right\|_2^2 + \frac{1}{n^t}\sum_{j=1}^{n^t}\left\|\mathbf{f}^t_{\text{mul}} - \mathbf{f}^t_{\text{uni}}\right\|_2^2\right) \tag{25}$$

where $\|\cdot\|_2$ denotes the Euclidean norm.

However, while SimKD efficiently reduces model complexity, it does not inherently ensure domain invariance of the distilled knowledge. A straightforward solution would be aligning the source and target domains within the student model, that is, enforcing cross-domain consistency between the student source features $\mathbf{f}^s_{\text{uni}}$ and the student target features $\mathbf{f}^t_{\text{uni}}$. However, since the student model is learnable, its alignment performance heavily relies on the quality of its feature extraction. If the student features are biased toward a particular

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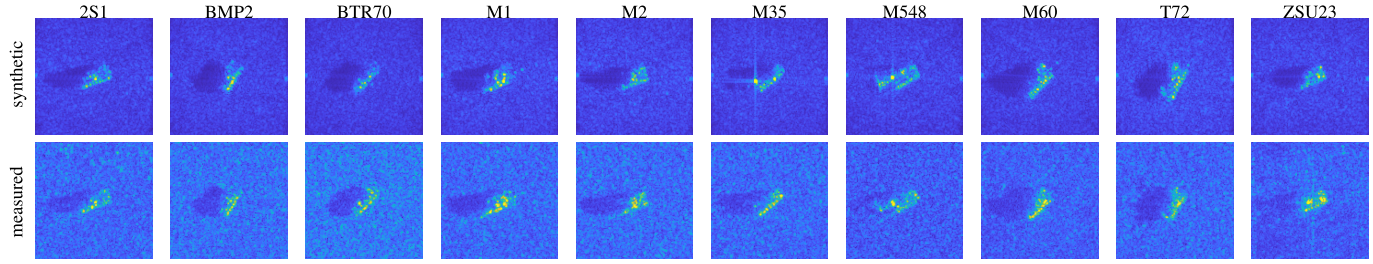


Fig. 16. Coupled synthetic and measured images in the SAMPLE dataset.

domain, the alignment may fail to capture domain-invariant patterns critical for robust cross-domain generalization.

To overcome this limitation, we propose a novel tri-path collaborative alignment strategy that introduces a teacher–student bidirectional cross-domain constraint, extending intrastudent alignment into a collaborative framework. The bidirectional cross-domain constraint uses the accurate multimodal features of the teacher model as anchors for domain alignment. By enforcing cross-domain consistency between the teacher source features $\mathbf{f}^{s}_{mul}$ and the student target features $\mathbf{f}^{t}_{uni}$, as well as between the teacher target features $\mathbf{f}^{t}_{mul}$ and the student source features $\mathbf{f}^{s}_{uni}$, the student is guided to learn domain-invariant representations that are robust to distribution changes, thereby reducing the risk of suboptimal alignment caused by the bias of the student features. In summary, the tri-path collaborative strategy overcomes the domain invariance limitation of SimKD, ensuring the distilled model achieves robust cross-domain recognition performance.

The domain alignment loss with the tri-path collaborative alignment strategy is defined as

$$\mathcal{L}_{da} = \mathcal{L}_{lmmd}\left(\mathbf{f}^{s}_{uni}, \mathbf{f}^{t}_{uni}\right) + \mathcal{L}_{lmmd}\left(\mathbf{f}^{s}_{mul}, \mathbf{f}^{t}_{uni}\right) + \mathcal{L}_{lmmd}\left(\mathbf{f}^{s}_{uni}, \mathbf{f}^{t}_{mul}\right). \tag{26}$$

In summary, the total loss function of the DCKD algorithm can be described as

$$\mathcal{L}_{uni} = \mathcal{L}_{kd} + \lambda_2 \cdot \mathcal{L}_{da} \tag{27}$$

where λ_2 is the weight coefficient. And to distinguish it from the multimodal model MFJA, the unimodal model obtained after distillation is named uni-MFJA.

V. Experiments

A. Dataset Description

The public SAMPLE dataset, comprising ten categories of ground vehicle SAR images, is used as the benchmark dataset to evaluate the cross-domain recognition performance of the proposed MFJA and the distilled uni-MFJA. It encompasses 1345 pairs of synthetic and measured SAR images within the range of depression angles from 14° to 17° and the range of azimuth angles from 10° to 80°. One pair of images from each category is illustrated in Fig. 16. The images are captured by an X-band SAR system operating in the HH polarization mode, featuring a spatial resolution of 0.3 × 0.3 m. Each image measures 128 × 128 pixels. The number of images in each category at each depression angle is shown in Table I.

TABLE I
Details of the SAMPLE Dataset

Category \ Depression Angle	Measured/Synthetic 14°	15°	16°	17°	Totals
2S1	0	66	50	58	174
BMP2	0	0	55	52	107
BTR70	0	0	43	49	92
M1	26	0	52	51	129
M2	23	0	52	53	128
M35	24	0	52	53	129
M548	23	0	52	53	128
M60	0	65	51	60	176
T72	0	0	56	52	108
ZSU23	0	66	50	58	174
Totals	96	197	513	539	1345

Currently, there are three mainstream experimental scenarios: two transductive scenarios and one inductive scenario. The transductive and inductive scenarios are two common learning scenarios in machine learning. The transductive learning scenario aims to leverage both labeled training samples and unlabeled test samples to directly infer the labels of these test samples, without prioritizing generalization to unseen samples. Conversely, the inductive learning scenario involves training a model solely on labeled training samples and then generalizing the learned knowledge to unseen samples. In the context of the domain adaptation task that is the focus of this article, the transductive learning scenario only involves two available datasets: the source domain training set and the target domain test set. Consequently, the target domain samples used for testing are also included in training. In contrast, the target domain is further divided into a target domain validation set and a target domain test set in the inductive scenario. The former is used as unlabeled target samples during training, while the latter is strictly isolated from the training process and used exclusively for evaluating recognition performance.

In the first transductive scenario, the source domain training set comprises synthetic images with depression angles from 14° to 16°, while the target domain test set consists solely of measured images with 17° depression angle. In the second transductive scenario, the source domain training set is composed of all the synthetic images and the target domain test set

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consists of all the measured images. While the synthetic and measured images exhibit a one-to-one correspondence, this inherent relationship is not available since the target domain is unlabeled during training. In the inductive scenario, all the synthetic images are considered as the source domain training set, while the measured images are divided into two parts: a target domain validation set consisting of images with depression angles from 14° to 16° and a target domain test set consisting images with 17° depression angle. For the sake of simplicity, these three scenarios are referred to as *Scenario I*, *Scenario II*, and *Scenario III* in the following discussion.

B. Experimental Setup

The subaperture decomposition number c is set to 3. In the PN-WGRL, the weight coefficient λ, the power β, and the temperate coefficient τ are set to 1, 0.25, and 0.07, respectively.

The multimodal MFJA is trained on batches of 128 samples over a total of 40 epochs. The adaptive moment estimation optimizer [44] is used for the optimization process. The learning rate and the weight decay factor are initialized to 10^{-3} and 10^{-5}, respectively. Since the LMMD loss is analogous to a softened pseudolabeling method, misclassified samples are able to seriously interfere with subdomain alignment when the recognition performance is relatively limited. According to [42], the weight of the LMMD loss is gradually increased from 0 to 1 throughout the whole training process, which is defined as

$$\lambda_1 = \frac{2}{1+\exp\left(-10\times\frac{\text{epoch_current}}{\text{epoch_total}}\right)} - 1 \tag{28}$$

where epoch_current and epoch_total denote the current epoch and the total number of epochs, respectively.

During the KD process, the total number of epochs is set to 30. According to [17], the stochastic gradient descent optimizer [45] with a Nesterov momentum of 0.9 and an initial learning rate of 10^{-2} is used. After 20 epochs of training, the learning rate is reduced to 10^{-3}. In the unimodal loss $\mathcal{L}_{\text{uni}}$, the weight of the domain alignment loss λ_2 is computed in the same way as shown in (28).

To ensure statistical reliability, each MFJA experiment is independently repeated 50 times. Accuracy from the last five epochs of each run is recorded, yielding a total of 250 scores. For each DCKD experiment, the ten MFJA models closest to the average accuracy are selected as teacher models. Each teacher model undergoes five independent distillation runs, with the accuracy from the last three epochs recorded, resulting in a total of 150 scores. For both MFJA and DCKD, the mean and standard deviation of the collected scores are reported as performance metrics.

C. Comparison Experiments

To validate the effectiveness of our proposed multimodal MFJA model and DCKD algorithm, a comparative performance analysis is conducted with a large number of algorithms in three experimental scenarios, including recent algorithms proposed for SAR images and advanced algorithms designed for optical images.

TABLE II
COMPARISON EXPERIMENTS IN SCENARIO I

Methods	Image	Accuracy
Lewis (2019) [5]	SAR	24.97%
Scarnati (2019) [46]	SAR	55.62%
Li (2020) [47]	Optical	$88.57 \pm 1.33\%$
Li (2021) [48]	Optical	$91.28 \pm 2.58\%$
Inkawhich (2021) [49]	SAR	$95.06 \pm 0.87\%$
Wang (2019) [50]	Optical	$96.33 \pm 0.36\%$
Sun (2023) [11]	SAR	$97.36 \pm 0.87\%$
VSFA (2023) [13]	SAR	$99.15 \pm 0.86\%$
uni-MFJA	SAR	$99.59 \pm 0.25\%$
MFJA	SAR	$\mathbf{99.59 \pm 0.44\%}$

TABLE III
COMPARISON EXPERIMENTS IN SCENARIO II

Methods	Image	Accuracy
Lewis (2020) [51]	SAR	$82.05 \pm 0.53\%$
Li (2021) [48]	Optical	$89.62 \pm 2.24\%$
Li (2020) [47]	Optical	$90.97 \pm 2.04\%$
Dong (2022) [52]	SAR	93.66%
Sun (2023) [11]	SAR	$96.46 \pm 1.14\%$
Melzer (2021) [53]	SAR	96.88%
Wang (2019) [50]	Optical	$96.95 \pm 0.41\%$
VSFA (2023) [13]	SAR	$98.18 \pm 0.97\%$
uni-MFJA	SAR	$99.25 \pm 0.15\%$
MFJA	SAR	$\mathbf{99.37 \pm 0.70\%}$

The recognition performance in three experimental scenarios is shown in Tables II–IV, respectively. The results indicate that the proposed MFJA achieves superior recognition performance in both the transductive scenarios. Meanwhile, in the inductive scenario, it achieves recognition performance comparable to the current SOTA methods. In fact, the majority of methods with relatively satisfactory results, including [11], [12], [13], [50], and our proposed MFJA, are dedicated to aligning both the global domain-level and local class-level distributions between the source and target domains, demonstrating the importance of local alignment in domain adaptation. Since the method in [50] is designed for optical images, it lacks the utilization of the domain knowledge inherent to SAR images. Similarly, although the method in [11] is designed for SAR images, it actually regards SAR images as optical images, thus failing to leverage SAR domain knowledge. Therefore, the performance of these two methods is suboptimal. In contrast, both our previously proposed VSFA and the method proposed in [12] incorporate SAR domain knowledge into deep learning and thus achieve better recognition performance.

As observed, the proposed MFJA outperforms the current SOTA method, VSFA, by 0.44% and 1.19% in two

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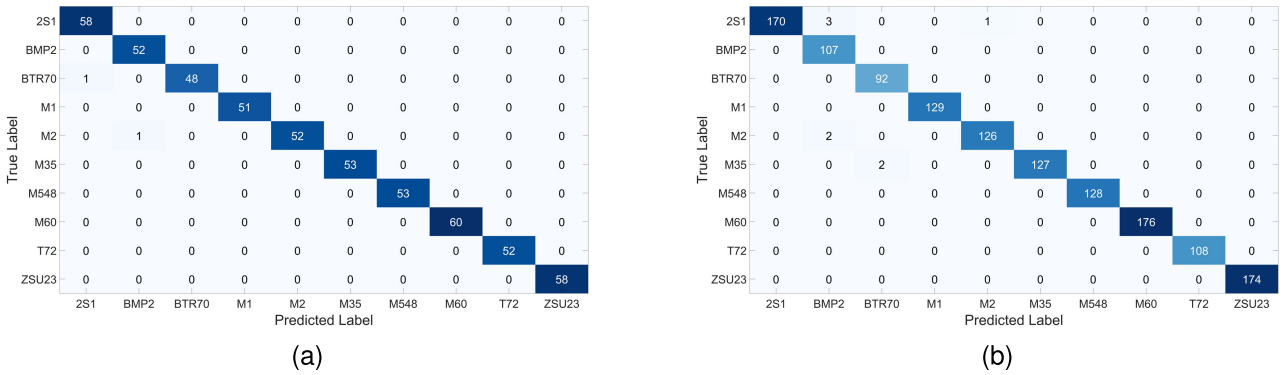


Fig. 17. Visualization of confusion matrices. (a) Scenario I. (b) Scenario II.

TABLE IV
COMPARISON EXPERIMENTS IN SCENARIO III

Methods	Image	Accuracy
Chen (2019) [54]	Optical	87.49%
Sharma (2021) [55]	Optical	89.91%
Lv (2022) [10]	SAR	90.72%
Saha (2022) [56]	Optical	91.47%
Dai (2020) [57]	Optical	92.73%
Chen (2022) [58]	SAR	93.69%
uni-MFJA	SAR	$96.19 \pm 1.48\%$
MFJA	SAR	$98.29 \pm 2.20\%$
Shi (2024) [12]	SAR	$98.44 \pm 0.34\%$
VSFA (2023) [13]	SAR	$\mathbf{98.80 \pm 1.00}\%$

transductive scenarios, but underperforms by 0.51% in the inductive scenario. We argue that the performance discrepancy is primarily due to the behavior of BN layers in MFJA and the distribution of the data itself. Although the BN layer accelerates training and improves model performance by smoothing loss landscapes [59], its standard implementation becomes suboptimal in domain adaptation tasks. Specifically, the statistics of BN are computed over mixed mini-batches containing both the source and target samples. The brute-force cross-domain sharing of statistics may distort feature distributions due to domain shifts [50]. The network architecture of VSFA does not contain BN layers and therefore is not limited by this issue, but at the cost of requiring more training epochs (150 versus 40 in MFJA). To test this hypothesis, we replaced the standard BN in MFJA with two plug-and-play variants: adaptive BN [60] and domain-specific BN [61]. Both the modifications significantly enhance generalization performance, boosting inductive accuracy to 99.49% $\pm$ 0.61% and 99.24% $\pm$ 0.84%, respectively. These improvements, however, are attributed to the mitigation of BN-related limitations and do not represent our core contribution.

In conclusion, the results of the comparison experiments validate the effectiveness of the proposed MFJA in transductive scenarios, while also indicating room for improvement in inductive scenarios. In addition, it is found that the distilled uni-MFJA is able to retain almost the original performance in transductive scenarios, while exhibiting a decline in performance by 2.10% in the inductive scenario. Nevertheless, the performance is still only worse than the method proposed in [12] and VSFA, outperforming all other methods. Further analyses are presented in Section V-F.

D. Visualization

In this section, the recognition performance of the proposed MFJA is further analyzed through visualization, using Scenarios I and II as illustrative examples. In particular, the model whose performance is closest to the average performance of 250 replicate experiments is chosen for visualization.

1) Confusion Matrix: The confusion matrix offers a comprehensive analysis of the classification results, facilitating the quantification of both accurate and inaccurate predictions for each class. As shown in Fig. 17, the results validate that the proposed MFJA is capable of accurately recognizing samples from the majority of categories, including eight categories in Scenario I and seven categories in Scenario II. In addition, the instances of category confusion mainly occur between BMP2 and M2 categories, as well as between BMP2 and 2S1 categories.

2) t-SNE Visualization: The t-SNE algorithm [62] is used to illustrate the distribution of images and the distribution of features for both the scenarios, as depicted in Fig. 18. Synthetic and measured images within the same category often occupy distinct regions in the feature space, while those from different categories may show significant overlap, indicating notable distribution differences between synthetic and measured images. In addition, multimodal features extracted from images exhibit strong intraclass compactness and interclass separability. Notably, the feature distributions of synthetic and measured images within the same category almost entirely overlap, demonstrating the effectiveness of the proposed MFJA in extracting domain-invariant features.

E. Ablation Experiments on MFJA

In this section, a series of ablation experiments are conducted in two transductive scenarios to analyze in detail the proposed MFJA.

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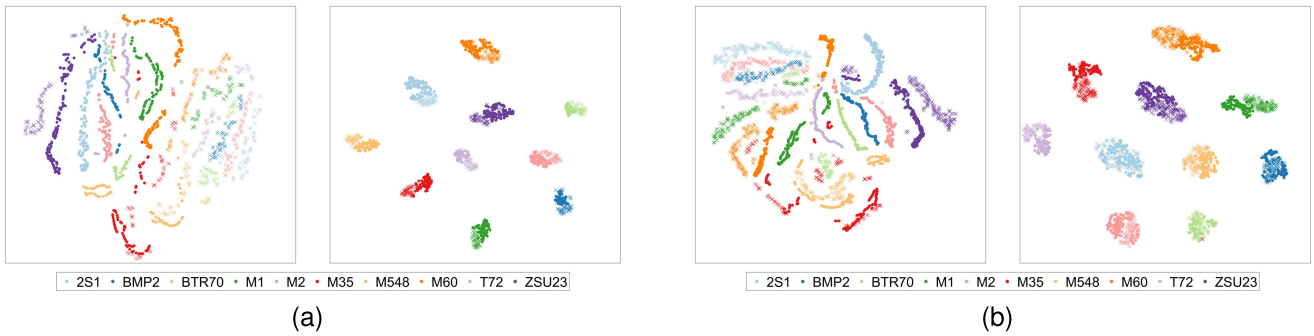


Fig. 18. Distribution visualization of images and features. (a) Scenario I. (b) Scenario II. (Left panel) Each figure depicts the distribution visualization of the original full-aperture images, while (right panel) illustrates the distribution visualization of the multimodal features extracted from the images by MFJA. The same color indicates the same category in synthetic images (represented by ·) and measured images (represented by ×).

TABLE V
ABLATION EXPERIMENT OF DIFFERENT COMPONENTS IN MFJA

SFFN	PA-GAT	GFFM	PN-WGRL	$\mathcal{L}_{lmmd}$	Accuracy (Scenario I / Scenario II)
✓†					$26.24 \pm 7.82\%$ / $28.02 \pm 6.91\%$
✓					$80.90 \pm 2.51\%$ / $78.78 \pm 1.77\%$
✓			✓	✓	$99.07 \pm 0.84\%$ / $98.81 \pm 1.06\%$
	✓				$72.58 \pm 2.47\%$ / $70.81 \pm 1.70\%$
	✓		✓	✓	$85.08 \pm 2.04\%$ / $87.79 \pm 1.44\%$
✓	✓		✓	✓	$99.21 \pm 0.82\%$ / $98.83 \pm 1.18\%$
✓	✓	✓		✓	$99.50 \pm 0.75\%$ / $98.60 \pm 1.21\%$
✓	✓	✓	✓	✓	$\mathbf{99.59 \pm 0.44\%}$ / $\mathbf{99.37 \pm 0.70\%}$

1) Analysis on Each Component: The multimodal MFJA can actually be divided into six components: the SFFN aimed at extracting visual features, the PA-GAT aimed at extracting scattering topological features, the GFFM aimed at fusing visual features and scattering topological features, the PN-WGRL aimed at global alignment (including the discriminative loss), the LMMD loss aimed at local alignment, and the classification loss. Given the necessity of the classification loss, it is not included in the ablation experiments. The ablation experiments for the other components are shown in Table V. The symbol † denotes that the corresponding experiment uses only source samples as input during training. In contrast, experiments without the symbol involve mixed mini-batches containing both the source and target samples. Since PA-GAT does not contain BN layers, its performance remains unaffected by the input type; therefore, no results with the symbol are reported for PA-GAT alone in Table V.

First, when the model is trained using only source samples, the vision-driven SFFN achieves a recognition rate of approximately 27% on target samples. This result reaffirms the substantial domain gap between synthetic and measured SAR images. The poor performance can be partially attributed to the presence of BN layers within the SFFN architecture. Training exclusively on source samples causes the statistics in BN layers to become specific to the source domain, thereby limiting their generalization capability in the target domain with a different data distribution. However, when mixed mini-batches containing both the source and target samples are used during training, the performance improves significantly—reaching approximately 80%—even without explicit domain adaptation constraints. Although standard BN layers are suboptimal for cross-domain tasks, they still contribute to domain alignment by mapping both the domains into an intermediate space with reduced distribution discrepancy, thus enhancing cross-domain recognition performance. In contrast, the topology-driven PA-GAT achieves a recognition accuracy of approximately 70%. This is due to its ability to capture local gradient and global structural information, which are relatively insensitive to statistical variations across domains. This observation supports the second advantage of incorporating domain knowledge: certain types of domain knowledge inherently exhibit stronger cross-domain invariance, which naturally mitigates the effects of domain shift.

Second, it is evident that each component contributes positively to overall performance. The introduction of global–local joint alignment leads to a significant enhancement in recognition accuracy for both the modalities, with an average improvement exceeding 15%, thereby demonstrating the effectiveness of domain adaptation for cross-domain recognition tasks. In the absence of GFFM, where visual and scattering topological features are concatenated for fusion, there is a notable lack of improvement in recognition performance compared with using visual features alone (from 99.07% to 99.21% in Scenario I and from 98.81% to 98.83% in Scenario II). Conversely, the incorporation of GFFM leads to more enhancement in the recognition performance (from 99.21% to 99.59% in Scenario I and from 98.83% to 99.37% in Scenario II). Notably, this result also indicates that multimodal feature

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TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT DOMAIN ADAPTATION METHODS

Scenario	I	II
LMMD	$99.50 \pm 0.75\%$	$98.60 \pm 1.21\%$
CORAL+LMMD	$99.42 \pm 0.77\%$	$98.82 \pm 1.04\%$
WGRL+LMMD	$99.56 \pm 0.56\%$	$98.99 \pm 1.03\%$
PN-WGRL+LMMD	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$

fusion outperforms unimodal representations, further validating the first advantage of incorporating domain knowledge: it enables more accurate target descriptions, which in turn facilitates more precise alignment under domain adaptation loss constraints. Furthermore, it is observed that the proposed PN-WGRL is able to assist the LMMD loss in aligning the distribution with greater precision, particularly in Scenario II.

2) Analysis on Global–Local Alignment: In this section, we further compare the improvement of several different global alignment methods on cross-domain recognition performance, with the LMMD loss serving as a benchmark. The detailed experimental results are shown in Table VI, where "CORAL+LMMD" denotes our previously proposed alignment strategy in VSFA, "PN-WGRL+LMMD" denotes our proposed alignment strategy in this article, and "WGRL+LMMD" indicates the replacement of PN-WGRL with the WGRL.

It is observed that a satisfactory recognition performance is achieved using the LMMD loss alone, primarily attributable to the enhanced discriminative capacity of the proposed multimodal feature extractor. For comparison, using the LMMD loss alone in VSFA yields only 97.01% recognition accuracy in Scenario I [13]. Accordingly, the incorporation of CORAL loss during the initial training stage of VSFA is able to mitigate the impact of misclassified samples on local alignment, thereby improving overall performance. In MFJA, however, the enhanced feature extractor enables accurate recognition for most samples, with only a small fraction of hard-to-align samples potentially disrupting local alignment. Consequently, the introduction of the CORAL loss, which is unable to mine hard samples, does not lead to performance improvements.

In addition, it is found that the recognition performance of Scenario I is significantly superior to that of Scenario II when the LMMD loss is used alone. Combined with the principle of LMMD (11), it can be assumed that the number of hard-to-align samples is limited in Scenario I, whereas a greater number are present in Scenario II. Consequently, the WGRL and the PN-WGRL, which possess the capacity to mine hard-to-align samples, only improve the recognition performance in Scenario II. Furthermore, the additional calibration of the weights allows our proposed PN-WGRL to further improve the performance based on WGRL.

3) Analysis on Subaperture Decomposition: In this section, we provide a detailed analysis of the effect of subaperture decomposition on recognition performance. As a benchmark for comparison, the recognition performance of various features derived from full-aperture SAR images is presented in Table VII. The results indicate that the visual modality significantly outperforms the scattering topological modality when used individually. Furthermore, the effectiveness of multimodal fusion in enhancing recognition performance is reaffirmed.

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT FULL-APERTURE FEATURES

Scenario	Visual	Topological	Visual + Topological
I	$97.81 \pm 1.56\%$	$85.08 \pm 2.04\%$	$99.01 \pm 1.02\%$
II	$98.13 \pm 1.17\%$	$87.79 \pm 1.44\%$	$99.00 \pm 0.72\%$

The performance comparison of different subaperture decomposition configurations is presented in Table VIII, where parameter c denotes the number of subapertures. To comprehensively evaluate the subaperture decomposition's effects, we systematically examine its application across different feature extraction branches, including both unimodal and multimodal configurations. The unimodal configurations consist of: 1) "Visual," which only uses visual features with subaperture decomposition and 2) "Topological," which only uses scattering topological features with subaperture decomposition. The multimodal configurations combine both the feature representations and include: 3) "Visual + Topological," where the subaperture decomposition is applied to both the branches simultaneously and 4) "Visual + Topological*" representing our optimized MFJA configuration that selectively applies decomposition only to the visual branch and not to the topological branch.

It can be observed that the recognition performance of various features extracted from a single subaperture image ($c = 1$) is inferior to that of the full-aperture benchmark. This phenomenon is to be expected, given that the single subaperture image contains only the central half of the spectrum present in the full-aperture image. When $c \geq 2$, the subaperture images contain all the spectrum of the full-aperture image. In this case, the incorporation of subaperture decomposition significantly enhances the recognition performance in all four different introduced approaches. The marginal effect of performance improvement tends to occur at $c = 3$. As c increases further, the decomposed subaperture images become increasingly similar and therefore provide very limited additional information.

In addition, as shown in the fourth and fifth rows of Table VIII, further introduction of subaperture decomposition into the scattering topological branch of MFJA, following its introduction into the visual branch, does not result in an improvement in recognition performance. We attribute this phenomenon to the dominance of visual modality. Therefore, subaperture decomposition is incorporated only into the visual branch, with the number c set to 3.

4) Analysis on SFFN: This section compares four distinct methods for fusing subaperture information. In addition to the three methods shown in Fig. 8, we also evaluate another commonly used method, which concatenates the subaperture images decomposed from a full-aperture image into a multichannel image [63], [64].

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TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT NUMBERS OF SUBAPERTURE DECOMPOSITION

Feature	c	1	2	3	4	5
Visual	I	$96.16 \pm 1.49\%$	$98.92 \pm 0.79\%$	$99.07 \pm 0.84\%$	$99.18 \pm 0.72\%$	$99.19 \pm 0.83\%$
	II	$97.84 \pm 1.15\%$	$98.73 \pm 1.16\%$	$98.81 \pm 1.06\%$	$98.98 \pm 1.04\%$	$98.93 \pm 1.03\%$
Topological	I	$75.26 \pm 2.34\%$	$85.19 \pm 1.78\%$	$86.45 \pm 2.02\%$	$86.44 \pm 2.25\%$	$86.84 \pm 2.16\%$
	II	$79.33 \pm 1.54\%$	$86.87 \pm 1.55\%$	$88.65 \pm 1.79\%$	$88.68 \pm 1.89\%$	$90.25 \pm 1.64\%$
Visual + Topological	I	$97.80 \pm 1.19\%$	$99.22 \pm 0.57\%$	$99.62 \pm 0.49\%$	$99.65 \pm 0.52\%$	$99.50 \pm 0.70\%$
	II	$98.37 \pm 0.78\%$	$99.23 \pm 0.80\%$	$99.25 \pm 0.70\%$	$99.21 \pm 0.78\%$	$99.30 \pm 0.67\%$
Visual + Topological*	I	$97.93 \pm 1.14\%$	$99.29 \pm 0.65\%$	$99.59 \pm 0.44\%$	$99.54 \pm 0.78\%$	$99.48 \pm 0.86\%$
	II	$98.49 \pm 0.71\%$	$99.22 \pm 0.94\%$	$99.37 \pm 0.70\%$	$99.25 \pm 0.71\%$	$99.30 \pm 0.72\%$

TABLE IX
PERFORMANCE COMPARISON OF DIFFERENT SUBAPERTURE FUSION METHODS

Scenario	I	II
Concatenation	$98.52 \pm 1.16\%$	$98.91 \pm 1.08\%$
Multi-Channel	$99.13 \pm 0.82\%$	$99.20 \pm 0.71\%$
LAFF	$99.48 \pm 0.61\%$	$99.29 \pm 0.81\%$
Average	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$

TABLE X
ABLATION EXPERIMENTS OF POSITION EMBEDDING

Model	Scenario	Without PE	With PE
PA-GAT	I	$56.72 \pm 5.64\%$	$85.08 \pm 2.04\%$
	II	$67.23 \pm 4.58\%$	$87.79 \pm 1.44\%$
MFJA	I	$99.23 \pm 1.30\%$	$99.59 \pm 0.44\%$
	II	$99.26 \pm 0.75\%$	$99.37 \pm 0.70\%$

As shown in Table IX, all three methods, with the exception of concatenation, improve recognition performance compared with extracting visual features from full-aperture images as presented in the fourth column of Table VII. In fact, the similarity between subaperture images is relatively high, indicating that there is a lot of redundant information between subaperture features. Consequently, the concatenation fusion method introduces unnecessary complexity, leading to reduced performance. Meanwhile, both the attention-based fusion and average fusion methods show comparable performance, which means that the importance between subaperture images is nearly equivalent. While the multichannel fusion method has the advantage of less computational complexity, its potential for performance improvement is relatively limited. Therefore, the average fusion method is ultimately selected for subaperture information fusion.

5) Analysis on PA-GAT: In this section, a series of ablation experiments are conducted to demonstrate the effectiveness of position embedding and linear branch integrated in PA-GAT. Specifically, PA-GAT and MFJA are regarded as benchmark models, respectively.

TABLE XI
ABLATION EXPERIMENTS OF LINEAR BRANCH

Model	Scenario	Without Linear Branch	With Linear Branch
PA-GAT	I	$81.33 \pm 2.54\%$	$85.08 \pm 2.04\%$
	II	$84.28 \pm 1.44\%$	$87.79 \pm 1.44\%$
MFJA	I	$99.39 \pm 0.62\%$	$99.59 \pm 0.44\%$
	II	$99.12 \pm 1.04\%$	$99.37 \pm 0.70\%$

As demonstrated in Table X, position embedding markedly enhances recognition performance when PA-GAT is used as the recognition model. Conversely, it provides only a modest improvement when applied to MFJA. This difference may be attributed to two factors. First, in PA-GAT, the position information introduced by position embeddings can be complemented with the scattering information captured by descriptors, leading to a notable boost in recognition performance. However, the CNN branch in MFJA is capable of implicitly learning the positional information to a certain extent [65], which diminishes the additional benefit of explicit position embeddings. Second, scattering topological features play only a complementary role in MFJA. Therefore, the enhancements made to PA-GAT translate into relatively limited improvements for MFJA (0.36% improvement in Scenario I and 0.11% improvement in Scenario II). Similarly, as shown in Table XI, the introduction of linear branch into PA-GAT significantly improves performance by 3.75% and 3.51%, whereas its introduction into MFJA results in only 0.20% and 0.25% performance improvements.

6) Analysis on GFFM: This section conducts a comparative analysis of the proposed GFFM alongside several fusion methods, including another GFM [66]. The GFM initiates the fusion process by concatenating features from different modalities to produce a combined feature vector. It then uses a linear layer in conjunction with the sigmoid function to learn a distinct weighted factor for each dimension. Ultimately, the features of different modalities are weighted summed, with the weighted factor differing for each dimension.

As presented in Table XII, the proposed GFFM exhibits superior performance compared with other methods. It is

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TABLE XII
PERFORMANCE COMPARISON OF DIFFERENT MULTIMODAL FEATURE FUSION METHODS

Scenario	I	II
LAFF	$98.86 \pm 1.01\%$	$98.79 \pm 1.16\%$
GFM	$98.98 \pm 0.93\%$	$98.88 \pm 1.17\%$
Average	$99.05 \pm 0.79\%$	$98.95 \pm 1.16\%$
Concatenation	$99.21 \pm 0.82\%$	$98.83 \pm 1.18\%$
GFFM	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$

TABLE XIII
ABLATION EXPERIMENT OF DIFFERENT COMPONENTS IN PN-WGRL

Power Transformation	Temperature-Scaled Softmax Normalization	Scenario I	Scenario II
		$99.56 \pm 0.56\%$	$98.99 \pm 1.03\%$
✓		$94.71 \pm 2.69\%$	$99.67 \pm 0.17\%$
	✓	$99.75 \pm 0.38\%$	$98.74 \pm 1.15\%$
✓	✓	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$

TABLE XIV
PARAMETRIC SENSITIVITY FOR POWER β

β	Scenario I	Scenario II
0.25	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$
0.50	$99.54 \pm 0.59\%$	$99.04 \pm 0.97\%$
0.75	$99.56 \pm 1.12\%$	$98.77 \pm 1.09\%$
1.00	$99.75 \pm 0.38\%$	$98.74 \pm 1.15\%$

TABLE XV
PARAMETRIC SENSITIVITY FOR TEMPERATURE COEFFICIENT τ

τ	Scenario I	Scenario II
0.07	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$
0.25	$99.53 \pm 0.72\%$	$98.82 \pm 1.10\%$
0.50	$99.59 \pm 0.56\%$	$98.84 \pm 1.04\%$
0.75	$99.60 \pm 0.65\%$	$98.57 \pm 1.20\%$
1.00	$99.67 \pm 0.49\%$	$98.33 \pm 1.21\%$

observed that summation-based methods generally yield lower performance than concatenation-based methods, as the latter better preserve feature diversity and mitigate information loss. This advantage is particularly beneficial for fusing multimodal features with inherent differences. In comparison to direct concatenation, the proposed GFFM is capable of more effectively mining complementary information and suppressing conflicting information between different modalities through its gating mechanism, leading to further improvements in recognition performance.

7) Analysis on PN-WGRL: In this section, a series of ablation and parameter sensitivity experiments are conducted on PN-WGRL. The results of the ablation experiments are presented in Table XIII. Note that the PN-WGRL will degenerate to the WGRL in the absence of power transformation and temperature-scaled softmax normalization. Since there are fewer hard samples in Scenario I, the power transformation dramatically increases the weights of easy samples, thereby shifting the focus of the model to the discrimination task, which results in a reduction in recognition performance. On the contrary, in Scenario II, where there are more hard samples, the power transformation is able to effectively mine hard samples, leading to improved recognition performance. The application of the softmax normalization alone has little impact on recognition performance. Its primary function is to further adjust the generally increased weights after power transformation, suppressing the weights of easy samples and emphasizing the weights of hard samples, which markedly improves recognition performance in Scenario I. Conversely, in Scenario II, the softmax normalization may suppress some relatively hard samples, thus slightly reducing recognition performance. In summary, we believe that PN-WGRL demonstrates more robust generalization across both the scenarios, making it suitable for a variety of scenarios.

The sensitivity experiments of the power β are shown in Table XIV, where the temperature coefficient τ is fixed at 0.07. It can be observed that the recognition performance in Scenario I remains largely unchanged across various powers, whereas it is significantly affected in Scenario II. This phenomenon is attributed to different numbers of hard samples in the two scenarios. As shown in Fig. 13(b), as the power β approaches 1, the amplifying effect of the power transformation on the weights of hard samples diminishes, leading to a decrease in recognition performance in Scenario II.

The sensitivity experiments of the temperature coefficient τ are shown in Table XV, where the power β is fixed at 0.25. Similarly, the temperature coefficient τ exerts a minor influence on Scenario I but a pronounced impact on Scenario II. The smaller the temperature coefficient τ, the sharper the output of the softmax function, which increases the weight assigned to hard samples, thereby further enhancing recognition performance. Ultimately, the values of power β and temperature coefficient τ are set to 0.25 and 0.07, respectively.

F. Ablation Experiments on DCKD

This section presents a series of ablation experiments to analyze in detail the proposed DCKD algorithm.

1) Analysis on Recognition Performance: It is worth noting that the designed DCKD algorithm is only applicable in inductive scenarios. This is because, in transductive scenarios, the multimodal model directly infers the labels of target domain samples, making KD unnecessary. Conversely, in inductive scenarios, the model is trained exclusively on source domain samples and target domain validation samples, with the aim of assessing its generalization capabilities on unseen target domain test samples. Therefore, distilling multimodal knowledge from the teacher model to the student model helps

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TABLE XVI
ABLATION EXPERIMENT OF DIFFERENT COMPONENTS IN DCKD

	Scenario I	Scenario II	Scenario III
Teacher	$99.59 \pm 0.44\%$	$99.37 \pm 0.70\%$	$98.29 \pm 2.20\%$
Student	$97.81 \pm 1.56\%$	$98.13 \pm 1.17\%$	$94.29 \pm 2.40\%$
SimKD	$99.39 \pm 0.34\%$	$98.96 \pm 0.39\%$	$94.69 \pm 1.87\%$
SimKD + Single-Path	$99.48 \pm 0.33\%$	$99.19 \pm 0.15\%$	$95.61 \pm 1.49\%$
SimKD + Tri-Path	$99.59 \pm 0.25\%$	$99.25 \pm 0.15\%$	$96.19 \pm 1.48\%$

accelerate inference on previously unseen samples. Nonetheless, the performances of both the transductive scenarios and one inductive scenario are still reported in ablation experiments, providing a more comprehensive analysis.

The results are presented in Table XVI. The terms "Teacher" and "Student" stand for the multimodal model and the visual model, respectively, both of which are trained independently of scratch. "SimKD" represents the distillation through the MSE loss. "Single-Path" represents the alignment within the student model. "Tri-Path" denotes our proposed tri-path collaborative alignment strategy, which further introduces the teacher–student bidirectional constraint.

First, it is evident that the recognition performance of the visual student model is significantly inferior to that of the multimodal teacher model, particularly in the inductive scenario. This observation demonstrates the effectiveness of the proposed multimodal feature extractor again and validates that the fusion of multimodal information is able to enhance the generalization capability of the model.

Second, it is observed that when the MSE loss is applied alone to constrain the KD process, the performances of both the transductive scenarios are substantially improved, almost approaching that of the teacher model. In contrast, the performance improvement is relatively limited in the inductive scenario. Since the student model reuses the classifier of the teacher model, in transductive scenarios, the features output by the student model only need to be matched with the features output by the teacher model to achieve satisfactory recognition performance. Indeed, there may be multiple mapping relationships in the feature space that satisfy feature matching. Without the imposition of additional constraints, the mapping learned by the student model may exhibit weak domain invariance. As a result, in the inductive scenario, KD guided by the MSE loss leads to only a more limited improvement in performance.

Third, it can be found that incorporating the domain adaptation constraint into the student model improves recognition performance across all three scenarios, with the most significant improvement observed in the inductive scenario. This phenomenon indicates that the domain invariance of features is lost during KD and validates the necessity of introducing domain adaptation constraints. Furthermore, the performance in all three scenarios is further improved following the replacement of the single domain alignment with the proposed tri-path collaborative alignment strategy, thereby substantiating its effectiveness.

TABLE XVII
ANALYSIS ON INFERENCE TIME IN MFJA

sub-aperture decomposition	graph construction	forward propagation	total
6.44ms	85.83ms	1.56ms	93.83ms

TABLE XVIII
ANALYSIS ON INFERENCE TIME IN UNI-MFJA

normalization	forward propagation	total
1.34ms	1.14ms	2.48ms (97.36% ↓)

TABLE XIX
ANALYSIS ON MODEL EFFICIENCY

	MACs (M)	Parameters (M)
MFJA	79.12	0.92
uni-MFJA	25.22 (68.12% ↓)	0.65 (29.35% ↓)

2) Analysis on Inference Efficiency: In this section, we evaluate the effectiveness of distillation in terms of inference efficiency on a computer (CPU: Inter Xeon Silver 4210R; RAM: 128 GB; and GPU: RTX 4090).

First, we measure the time taken by the multimodal model MFJA and the visual model uni-MFJA to recognize a new complex-valued full-aperture SAR image during inference. In MFJA, both subaperture decomposition and scattering topological graph construction are required before feeding the image into the network. In contrast, only the quarter power magnitude normalization is necessary in uni-MFJA. The results are depicted in Tables XVII and XVIII. It is evident that the preprocessing stage in MFJA is considerably more time-consuming than forward propagation of the network, particularly the scattering topological graph construction, which accounts for over 91.47% of the total inference time. This restricts the potential for applying the MFJA model in large-scale fast inference scenarios to some extent. The avoidance of time-consuming preprocessing in uni-MFJA allows the inference time to be reduced by 97.36%, reaching just 2.48 ms, which enables the network to predict over 400 images per second.

The comparison of inference times clearly demonstrates the computational complexity introduced by preprocessing algorithms. However, the forward propagation times of the models before and after distillation are relatively similar. To more comprehensively illustrate the optimization effect brought about by distillation, we further compare the number of MACs and parameters for both the models. As shown in Table XIX, the designed DCKD algorithm reduces the number of MACs and parameters of the model by 68.12% and 29.35%, respectively. The reduction in parameters is mainly attributed to the removal of the topological branch, while the decrease in MACs is primarily due to the substitution of the visual branch's input, transitioning from subaperture sequences to individual full-aperture images. As demonstrated in Table XVI, the performance of uni-MFJA exhibits a mere

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2.10% decrease, which is still superior to most of the algorithms compared, verifying the effectiveness of the proposed DCKD algorithm.

G. Limitations and Future Directions

Despite the satisfactory performance achieved, it is important to acknowledge that there are still several limitations in our methodology. We believe that these limitations primarily lie in three areas.

1) *Manual Hyperparameter Tuning*: The proposed PN-WGRL layer requires manual configuration of two critical hyperparameters (power β and temperature τ) that may impact cross-domain recognition performance. While we have provided and analyzed optimized settings under current experimental scenarios based on sensitivity experiments, hyperparameter optimization remains an open challenge. Future work could explore automated hyperparameter optimization methods or design more robust gradient weighting strategies.
2) *Two-Stage Pipeline:* The DCKD algorithm, designed to accelerate inference, transforms the original training process into a two-stage pipeline. This may introduce inherent teacher-to-student error propagation, thus affecting recognition performance. One promising direction for improvement would be to integrate multimodal feature fusion, domain adaptation, and KD into a unified single-stage framework.
3) *Generalization Performance:* While MFJA achieves the best performance in two transductive scenarios, it still shows limited generalization ability in the inductive scenario. We attribute this limitation primarily to the statistical shift in BN layers. Future research could explore dedicated strategies to further enhance the model's generalization capability.

VI. Conclusion

This article presents MFJA, a novel unsupervised domain adaptation framework for cross-domain recognition tasks in SAR ATR. The key innovations are threefold: 1) a comprehensive multimodal feature extractor for SAR images, encompassing subaperture feature extraction, scattering topological feature extraction, and gated feature fusion; 2) a PN-WGRL layer based on hard sample mining to reduce the interference of misclassified samples on local domain alignment; and 3) a DCKD algorithm with a tri-path collaborative alignment strategy that reduces model complexity while retaining both discriminative power and domain invariance of knowledge. The results from comparison and ablation experiments demonstrate that MFJA exhibits superior performance in most scenarios. In addition, the DCKD algorithm significantly improves the cross-domain recognition performance of the student model through its collaborative alignment strategy.

Our research reveals several insights about SAR domain adaptation: first, SAR-related domain knowledge is beneficial for cross-domain recognition; second, promoting local alignment facilitates more accurate cross-domain recognition; and third, the cross-domain invariance of knowledge is often diminished during the distillation process.

Despite the promising results, there is still room for improvement. Further research will be directed at refining the model to improve robustness, efficiency, and generalization across a broader range of practical applications.

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